COSKELETON FUNCTORS EXIST OVER A FINITELY COMPLETE CATEGORY AND ARE GIVEN BY THE LIMIT FORMULA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the mapping property is verified completely in both directions; the phrase 'it is easy to see that these morphisms form a morphism sk_m V -> U' (source lines 2481-2483) refers to assembling the components gamma(id_[k]) according to the definition of a morphism of m-truncated simplicial objects, which is precisely the data exhibited; functoriality of U |-> cosk_m U in U comes from the cited functoriality of limits, as stated in the lemma.

Editorial note. The label lemma-existence-cosk names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Limits over diagram categories and their functoriality. Categories chapter, \label{categories-lemma-functorial-limit}. Line 2464: the map cosk_mU(varphi) comes from the identification U(n') circ varphi-bar = U(n) via functoriality of limits.
- Truncated simplicial objects and the skeleton functor. Same chapter, the definition of sk_m and m-truncated simplicial objects. Lines 2480-2483: the assembled gamma(id_[k]) form a morphism sk_m V -> U.
- Finite limits exist in C. Assumption of the lemma. Line 2466: the limits over the finite categories (Delta/[n])_{<=m} exist.

Source reference. Original lemma 002L: sources/stacks-chow/tags.txt, source lines 102–102. Prerequisite pointer invoked at proof line(s) 2463; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If the category $\mathcal{C}$ has finite limits, then $cosk_m$ functors exist for all m. Moreover, for any m-truncated simplicial object U the simplicial object $cosk_m U$ is described by the formula*

$$(cosk_m U)_n = \lim_{(\Delta/[n])^{opp}_{\leq m}} U(n)$$

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

and for $\varphi : [n] \to [n']$ the map $cosk_m U(\varphi)$ comes from the identification $U(n') \circ \overline{\varphi} = U(n)$ above via Categories, Lemma 002L.

Proof. During the proof of this lemma we denote $\mathrm{cosk}_m U$ the simplicial object with $(\mathrm{cosk}_m U)_n$ equal to $\lim_{(\Delta/[n])^{opp}_{\leq m}} U(n)$. We will conclude at the end of the proof that it does satisfy the required mapping property.

Suppose that V is a simplicial object. A morphism $\gamma : V \to \mathrm{cosk}_m U$ is given by a sequence of morphisms $\gamma_n : V_n \to (\mathrm{cosk}_m U)_n$. By definition of a limit, this is given by a collection of morphisms $\gamma(\alpha) : V_n \to U_k$ where α ranges over all $\alpha : [k] \to [n]$ with $k \leq m$. These morphisms then also satisfy the rules that

$$\begin{array}{ccc} V_n & \xrightarrow[\gamma(\alpha)]{} & U_k \\ {\scriptstyle V(\varphi)}\uparrow & & \uparrow{\scriptstyle U(\psi)} \\ V_{n'} & \xrightarrow{\gamma(\alpha')} & U_{k'} \end{array}$$

are commutative, given any $0 \leq k, k' \leq m$, $0 \leq n, n'$ and any $\psi : [k] \to [k']$, $\varphi : [n] \to [n']$, $\alpha : [k] \to [n]$ and $\alpha' : [k'] \to [n']$ in Δ such that $\varphi \circ \alpha = \alpha' \circ \psi$. Taking $n = k = k'$, $\varphi = \alpha'$, and $\alpha = \psi = \mathrm{id}_{[k]}$ we deduce that $\gamma(\alpha') = \gamma(\mathrm{id}_{[k]}) \circ V(\alpha')$. In other words, the morphisms $\gamma(\mathrm{id}_{[k]})$, $k \leq m$ determine the morphism γ. And it is easy to see that these morphisms form a morphism $\mathrm{sk}_m V \to U$.

Conversely, given a morphism $\gamma : \mathrm{sk}_m V \to U$, we obtain a family of morphisms $\gamma(\alpha)$ where α ranges over all $\alpha : [k] \to [n]$ with $k \leq m$ by setting $\gamma(\alpha) = \gamma(\mathrm{id}_{[k]}) \circ V(\alpha)$. These morphisms satisfy all the displayed commutativity restraints pictured above, and hence give rise to a morphism $V \to \mathrm{cosk}_m U$. $\square$